

CMOS Technology: Where We Are and How We Got Here

KKKC6353 Nanoelectronic Devices · Lecture 1, Part 1 of 3

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The argument in three steps

1

30 years of free scaling

Shrinking transistors made them denser, faster and more efficient at once. This was a specific recipe: **Dennard scaling**.

2

Around 2003 it failed

V_T could not fall further without the transistor **leaking when it should be off**. A device-physics limit.

3

Every node since is an invention

Strained Si, high-k/metal gate, FinFET, GAA nanosheets, backside power, CFET: ways to keep scaling **without** the recipe.

How Lecture 1 is organised

Part 1 · CMOS technology

What the industry built, why classical scaling stopped working, and what replaced it.

Part 2 · Device physics

The physics you need, stated once in one notation, ending where that physics breaks down.

Part 3 · Short-channel effects

Why small transistors misbehave, and what is done about it.

1

From one transistor to the integrated circuit

In nanoelectronics, the concept is rarely the bottleneck.

The point-contact transistor, December 1947

- **Bardeen and Brattain**, in Shockley's group at Bell Labs
- First worked on **16 Dec 1947**; shown to management on 23 Dec
- Shockley then devised the manufacturable **junction** transistor
- Nobel Prize in Physics, **1956**
- But the field-effect idea was patented by **Lilienfeld in 1925**: about 40 years of materials processing separated idea from device



Bell Laboratories photograph; reproduced from W. A. Doolittle, Georgia Tech lecture slides (2011).

From Bell Labs to Silicon Valley

Why the lineage matters: it explains how a Fairchild observation about **manufacturing cost** became the organising principle of an industry.

1

1955–56

Shockley founds **Shockley Semiconductor** (funded by Beckman) in Mountain View

2

1957

Eight of his recruits, including **Moore** and **Noyce**, leave to found **Fairchild**

3

1965

Moore, Fairchild's research director, writes the article that becomes **Moore's law**

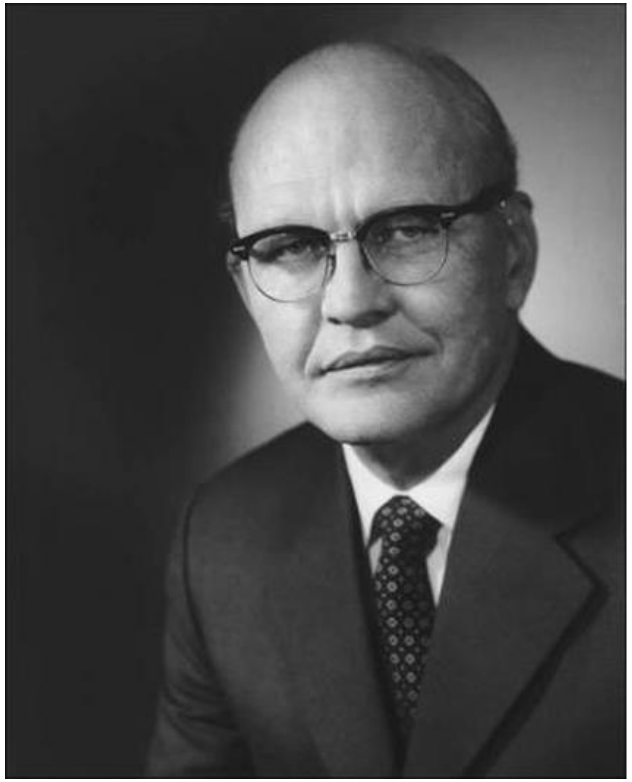
4

1968

Moore and Noyce leave Fairchild to found **Intel**

The integrated circuit, 1958–59

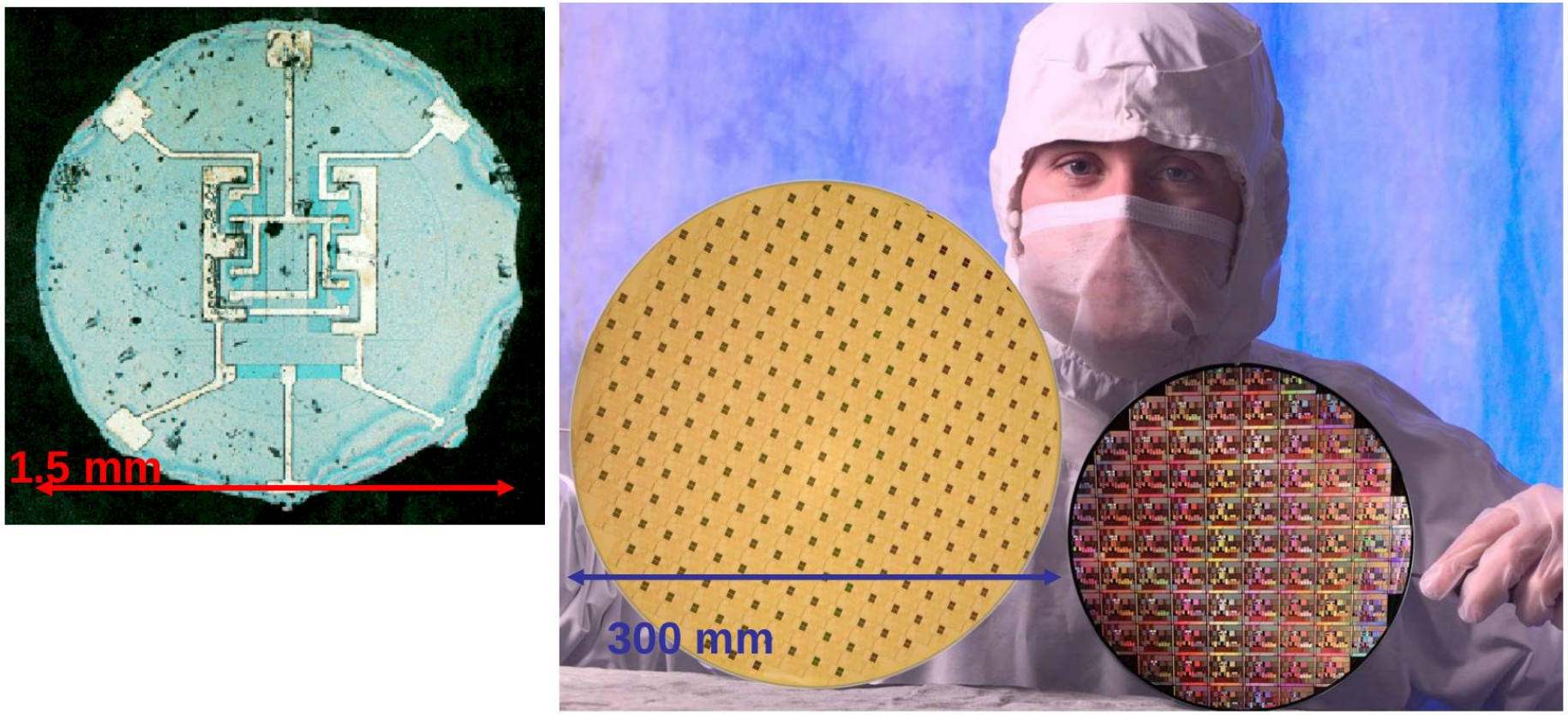
- **Jack Kilby** (Texas Instruments, 1958): several components on one piece of **germanium**
- **Robert Noyce** (Fairchild), independently: a **silicon** version using the **planar process**
- The planar process proved manufacturable: it is the ancestor of everything built since
- Kilby: Nobel Prize 2000. Noyce had died in 1990; the prize is not awarded posthumously





Left: Jack Kilby. Right: Grove, Noyce and Moore with the Intel 8080 layout. Photographs: Texas Instruments, Intel; reproduced from S. Levkov, NJIT ECE 271 slides.

Fifty years of the planar process



Left: Fairchild's first planar IC (1961), about 1.5 mm across. Right: 200 mm and 300 mm wafers. Sources: Smithsonian Chip Collection, IBM; reproduced from W. A. Doolittle, Georgia Tech lecture slides (2011).

The same printing idea, applied to ever larger wafers and ever smaller features.

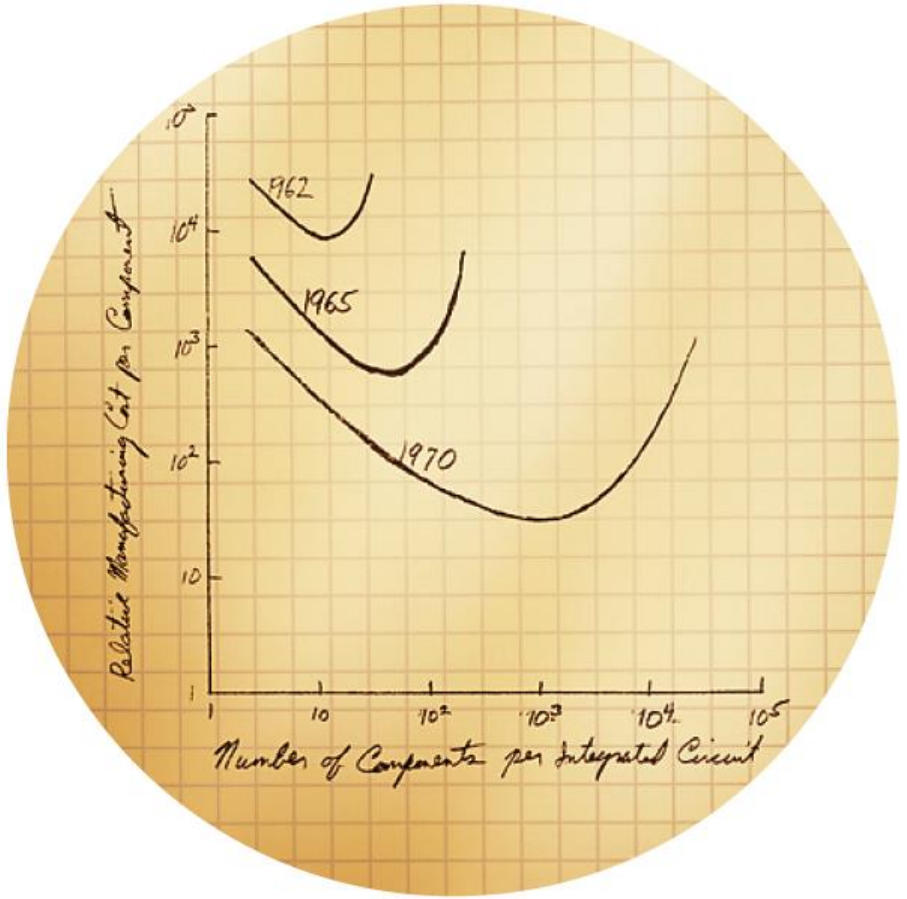
2

Moore's law: what it actually said

An economic observation, not a law of physics.

The 1965 paper

- Four-page article in *Electronics*, April 1965, by Fairchild's research director
- Components per IC had doubled **every year** since 1958; he expected a decade more
- **1975**: revised to a doubling **every two years**
- The axis is **relative cost per component**: each curve has a cost-optimal complexity, and that minimum moves right and down each year



G. E. Moore, Electronics 38(8), 1965. Image courtesy of Intel.

Three things students get wrong

1

It is economics, not physics

The quantity minimised is **cost per component**. It held for 60 years because an industry organised its investment to make it hold.

2

It is not 18 months

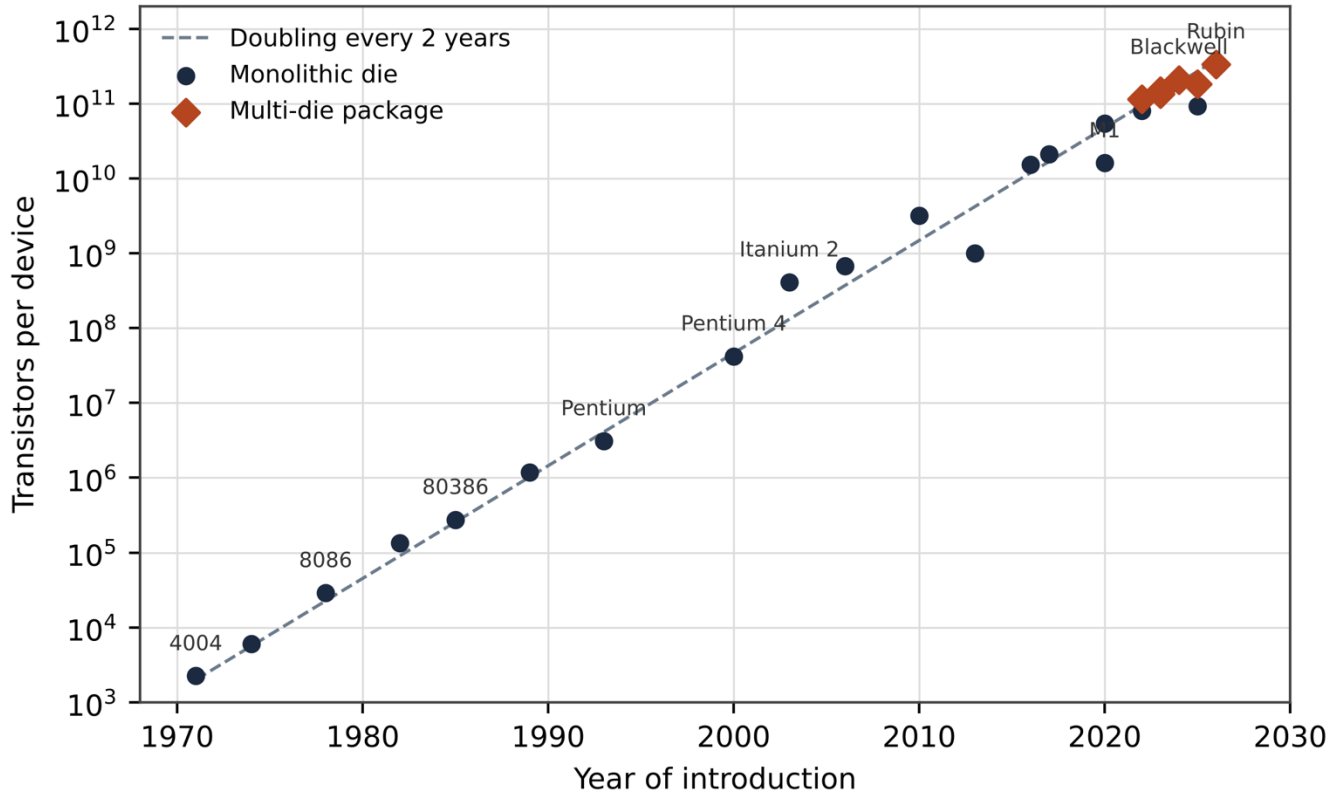
Moore said 1 year (1965), then **2 years** (1975). 18 months is a later paraphrase, usually attributed to David House at Intel.

3

Count is not performance

Transistor count and performance tracked each other until about **2004**, then separated completely. Section 4 is about why.

The trend through 2026

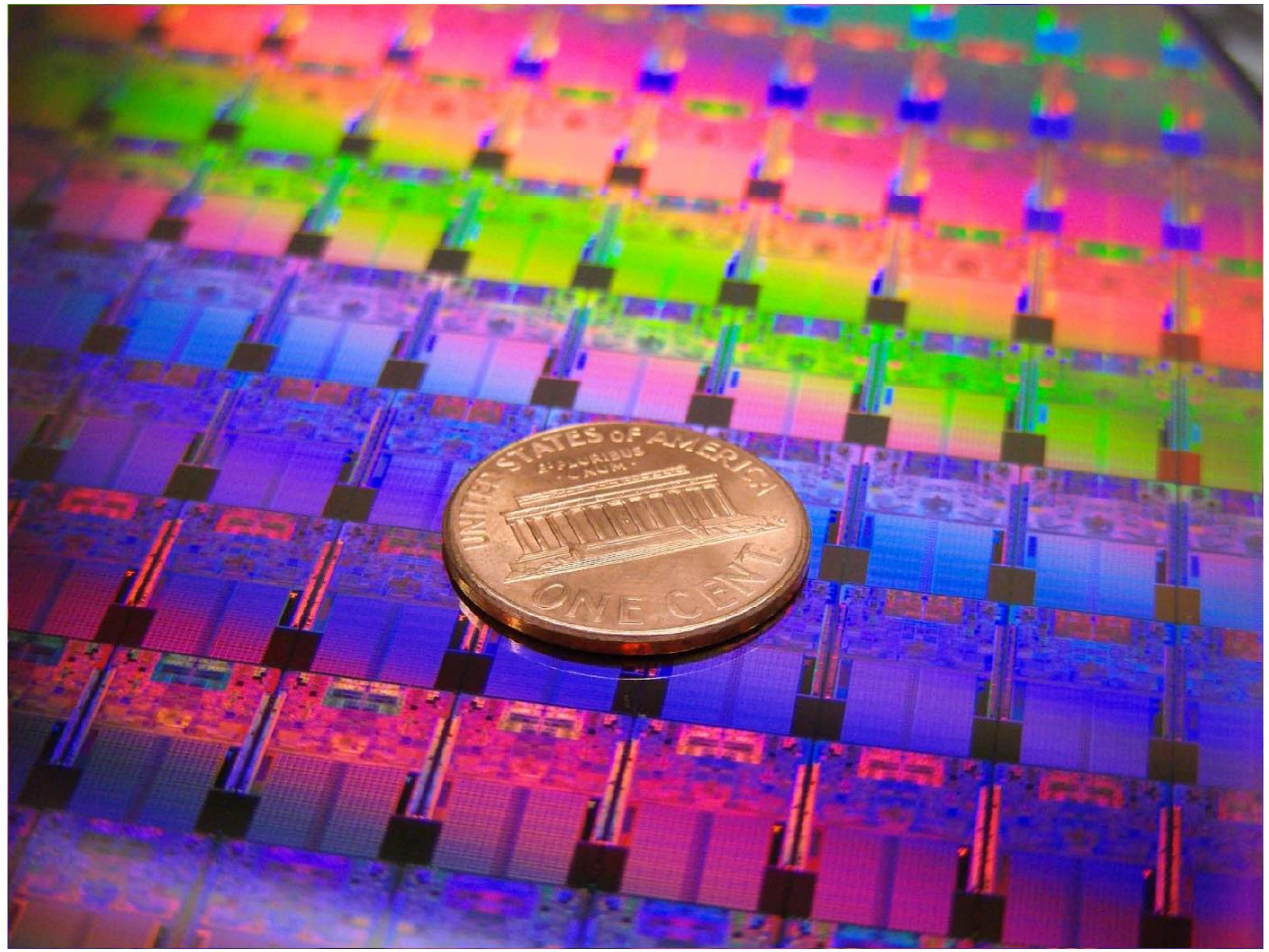


Data: Wikipedia, Transistor count (accessed October 2026).

The trend continues, but the top points are **multi-die packages** (Blackwell 208 bn, Rubin 336 bn). Does the law survive if you count monolithic dies only?

What one wafer holds

- Each rectangle is one die
- Each die: **10^8–10^9 transistors** (2008)
- Today's largest monolithic dies: **~10^{11}**
- Cost per transistor fell because the wafer cost was shared over ever more devices



300 mm wafer of Intel 45 nm processors (2008) with a one-cent coin. Image: Intel; reproduced from W. A. Doolittle, Georgia Tech lecture slides (2011).

3

Dennard scaling: the engine underneath

Moore's law says what happened. Dennard scaling says why it was worth doing.

Constant-field scaling (Dennard et al., IBM, 1974)

Quantity	Scales as	$\kappa = 1.4$
Device area	$1/\kappa^2$	halved
Gate capacitance $C = \varepsilon_{ox} WL/t_{ox}$	$1/\kappa$	× 0.71
Drive current I_D	$1/\kappa$	× 0.71
Gate delay $\tau \sim CV/I$	$1/\kappa$	× 0.71 (faster)
Power per device $P = IV$	$1/\kappa^2$	halved
Power density P/area	**constant**	**unchanged**
Electric field	constant	unchanged

Divide every dimension, V_{DD} and V_T by κ; multiply N_A by κ. Twice the transistors, each 40% faster, and **the chip gets no hotter**.

How the industry actually scaled

1970s–80s: constant voltage

V_{DD} held at **5 V** for logic compatibility, so the internal field **rose by κ** every generation.

Around 1980: the LDD

Rising field caused hot-carrier damage; the **lightly doped drain** was introduced to spread the drain field.

1990s–2000s: constant field

V_{DD} finally fell, **5 → 3.3 → 2.5 V …**, close to Dennard's recipe, until it stalled near **1 V** in the mid-2000s.

4

The end of Dennard scaling

A device-physics failure. Every topic in Part 3 hangs off it.

The chain of failure

1 V_{DD} must fall

Constant-field scaling needs a lower supply every node.

2 V_T must follow

Otherwise the overdrive $V_{DD} - V_T$ collapses, and drive current with it.

3 Subthreshold slope is bounded

Off-current falls exponentially with gate voltage, at best **60 mV/decade** at 300 K.

4 Each 60 mV costs 10×

Every 60 mV off V_T multiplies **every transistor's** leakage by ten.

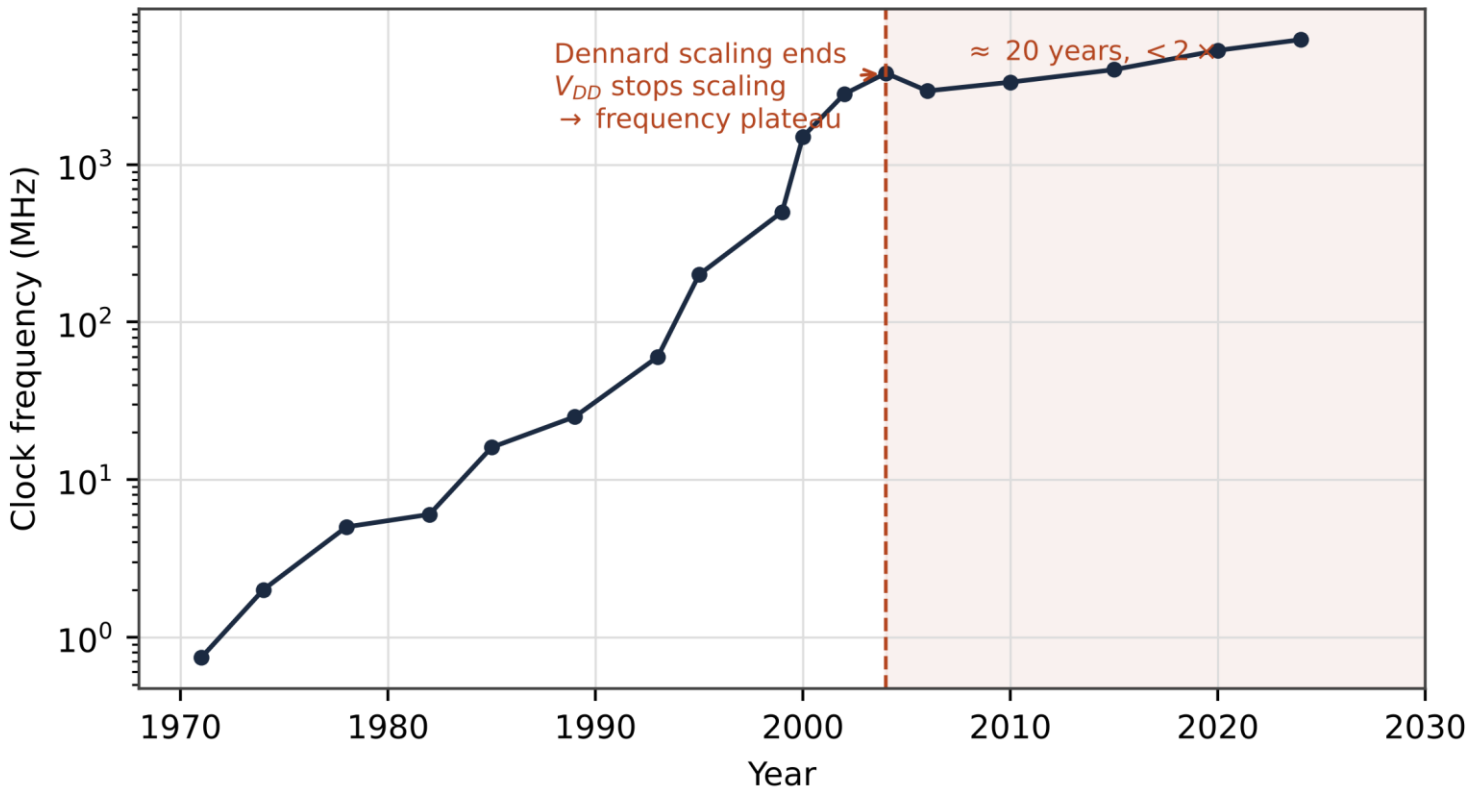
5 Leakage meets switching power

By the early 2000s, with 10^8 transistors per die, static power rivalled dynamic power.

6 Scaling stops

V_T stops, so V_{DD} stops, so **power density rises**.

What it looked like from outside: the clock stopped

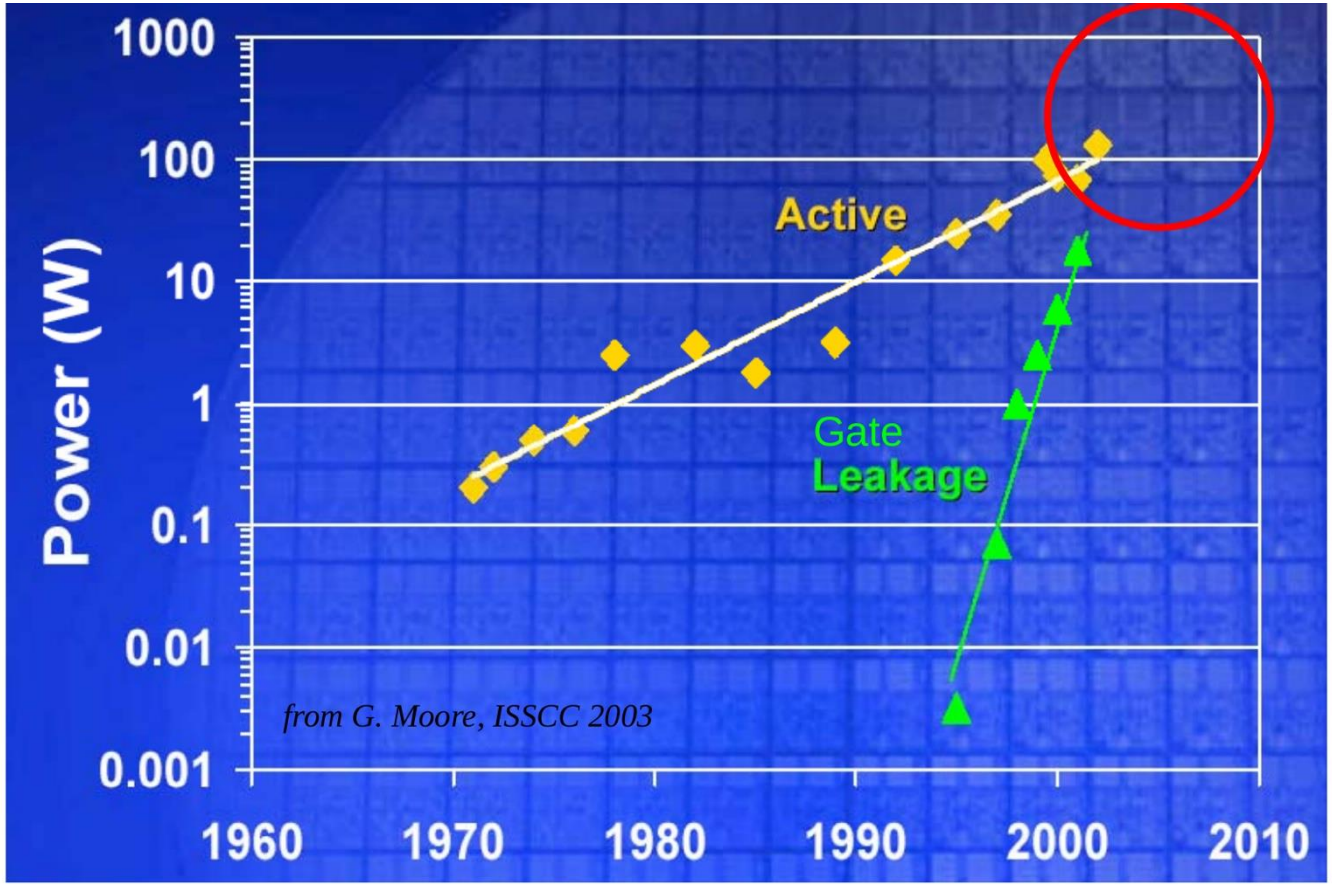


Representative maximum clocks at introduction; recent values are single-core boost clocks.

Frequency stalled near **4 GHz**. The transistors went into **parallelism**: multicore (2005), GPUs, then AI accelerators.

Leakage catches up with switching power

- Gate leakage was rising **far faster** than switching power
- On course to overtake it within a few years
- The response: **high-k / metal gate** (2007)
- **Dark silicon**: you can build more transistors than you can afford to switch on at once



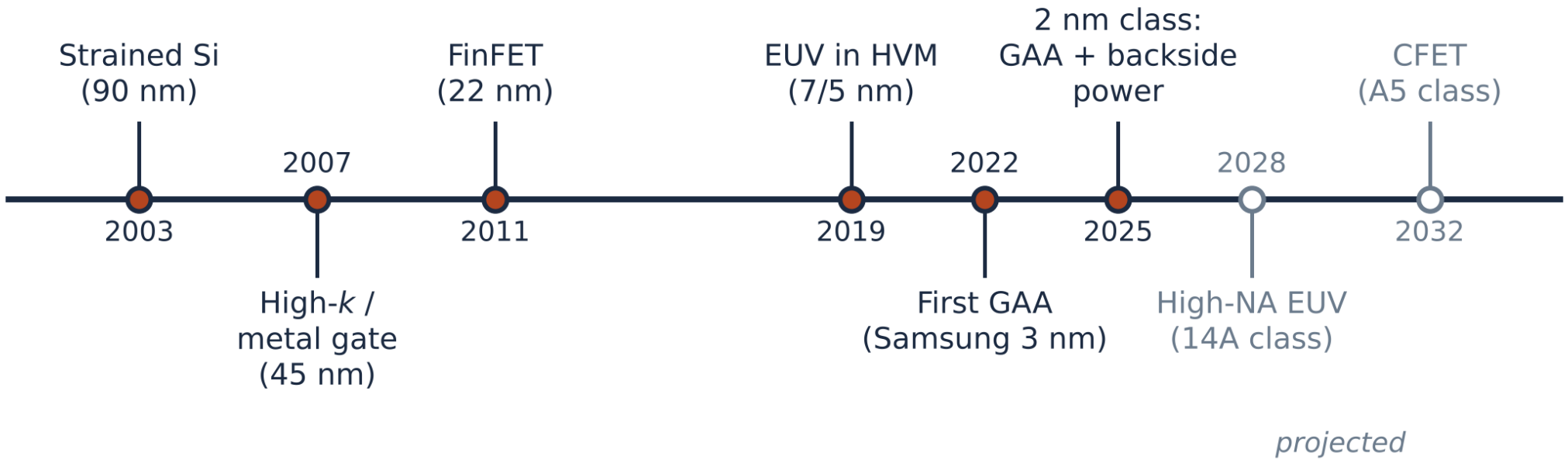
Active and gate-leakage power of microprocessors. Source: G. E. Moore, 'No exponential is forever', ISSCC 2003; reproduced from W. A. Doolittle, Georgia Tech lecture slides (2011).

5

What the industry did instead

With the free lunch gone, every node needed an invention.

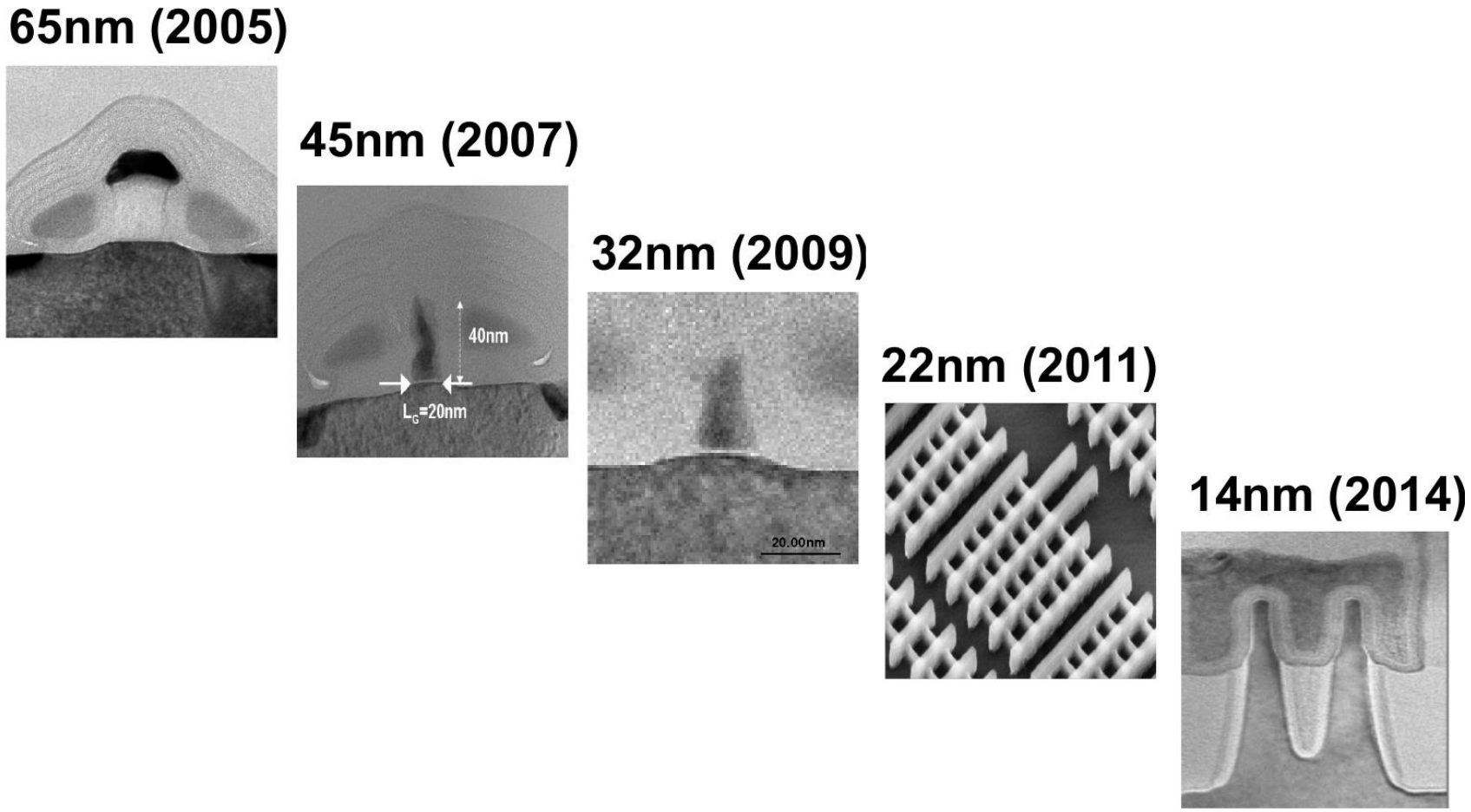
Twenty years of inventions



Dates before 2027 are volume-production dates; later entries are projections.

This sequence is also a preview of the rest of the module.

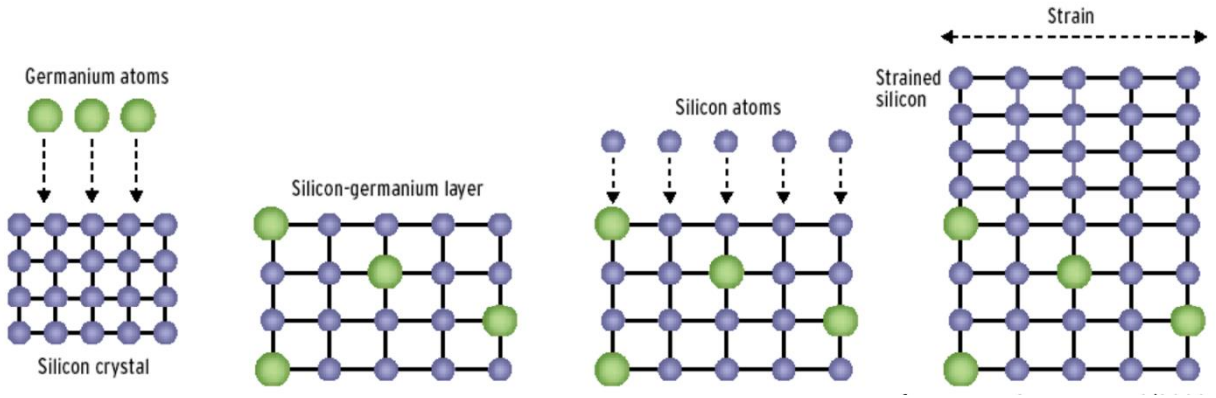
Intel, 65 nm to 14 nm: the planar device gives way to fins



Transistor cross-sections, one per generation, 2005–2014. Source: Intel; reproduced from ETH Zürich lecture slides.

Strained silicon (90 nm, 2003)

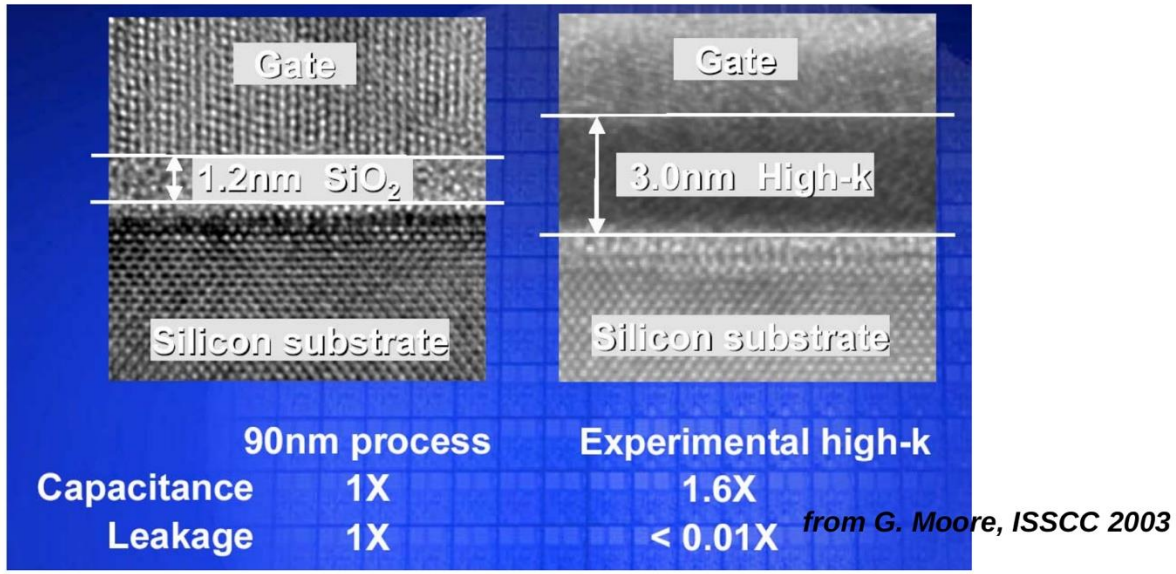
- Strain from SiGe layers, then stressor films and embedded SiGe source/drain
- **Mechanism:** strain lifts the conduction-valley degeneracy and warps the valence band
  - lower **transport effective mass**
  - less **intervalley scattering**
- Mobility rises because the band **shape** changes, **not** because the gap shrinks



Si grown on relaxed SiGe is stretched to the SiGe lattice. Source: IEEE Spectrum, Oct 2002; reproduced from W. A. Doolittle, Georgia Tech lecture slides (2011).

High-k dielectric and metal gate (45 nm, 2007)

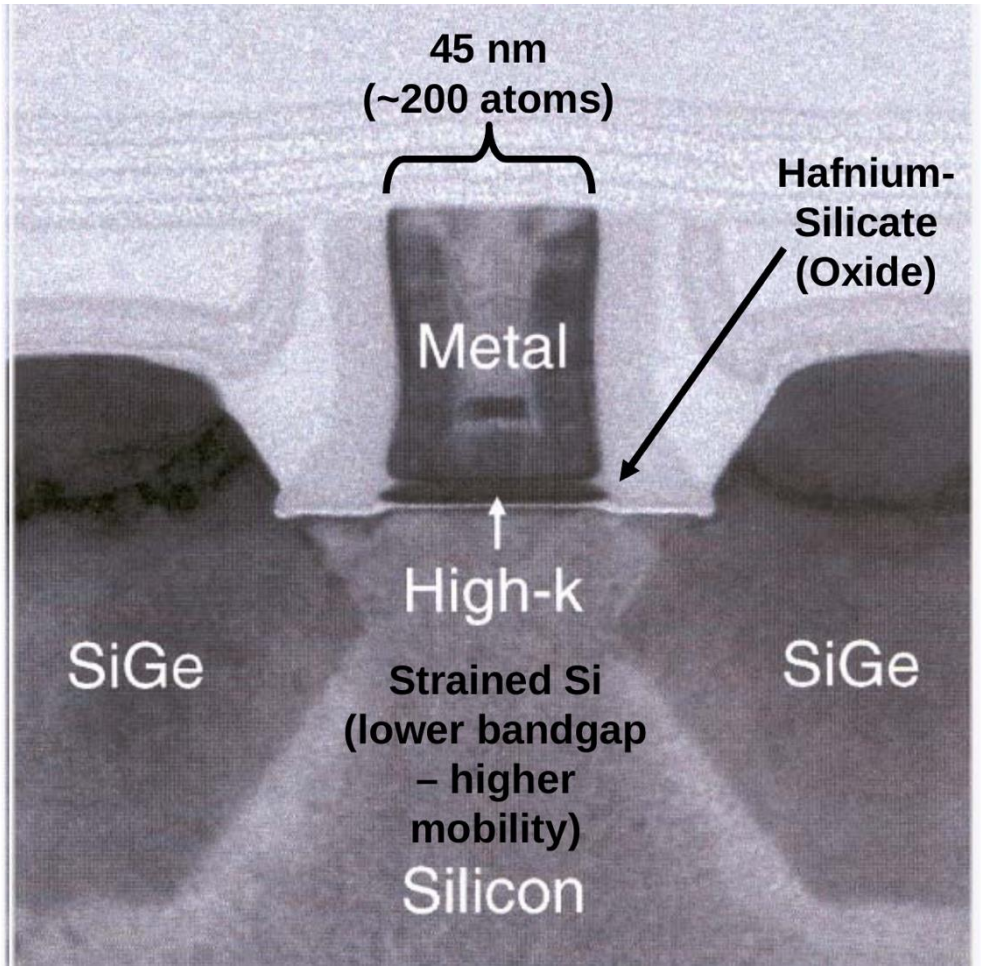
- By 90 nm, SiO_2 was ~**1 nm**: electrons tunnelled straight through
- Keep the **electrical** thickness, raise the **physical** thickness: EOT = t × 3.9 / k
- **HfO_2**-based, k ≈ 20–25: same C at 4–5× the thickness
- Poly-Si gates fail on high-k (Fermi-level pinning, phonon scattering), so **metal gates** returned



1.2 nm SiO_2 (90 nm process) beside 3.0 nm high-k: 1.6× the capacitance at under 1/100 of the leakage. Source: G. E. Moore, ISSCC 2003; reproduced from W. A. Doolittle, Georgia Tech lecture slides (2011).

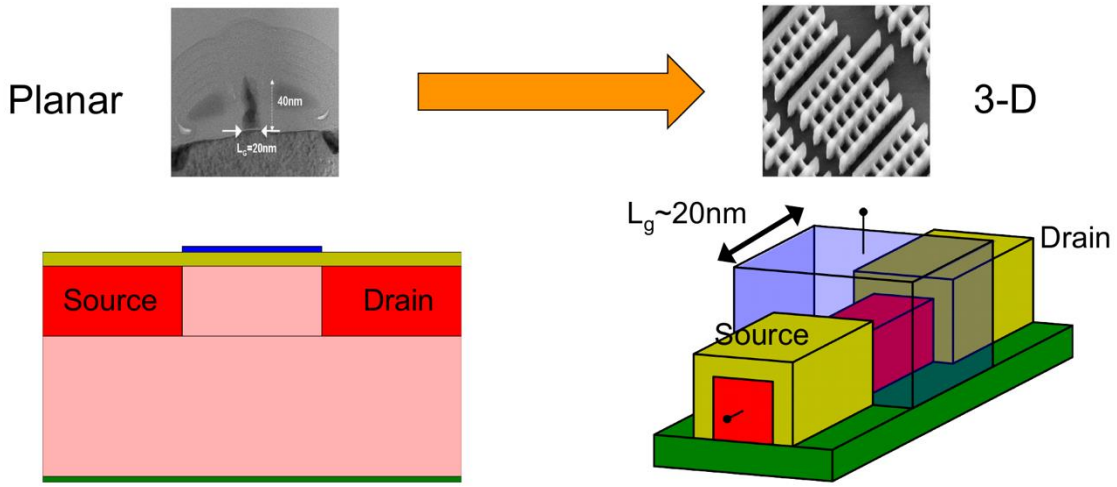
The 45 nm transistor as shipped

- Metal gate on a hafnium-based dielectric
- **Embedded SiGe** source/drain compresses the channel: strain for holes
- Read the label critically: 'lower bandgap → higher mobility' is the **wrong** mechanism
- The gain comes from the change in **band curvature** (previous slide)

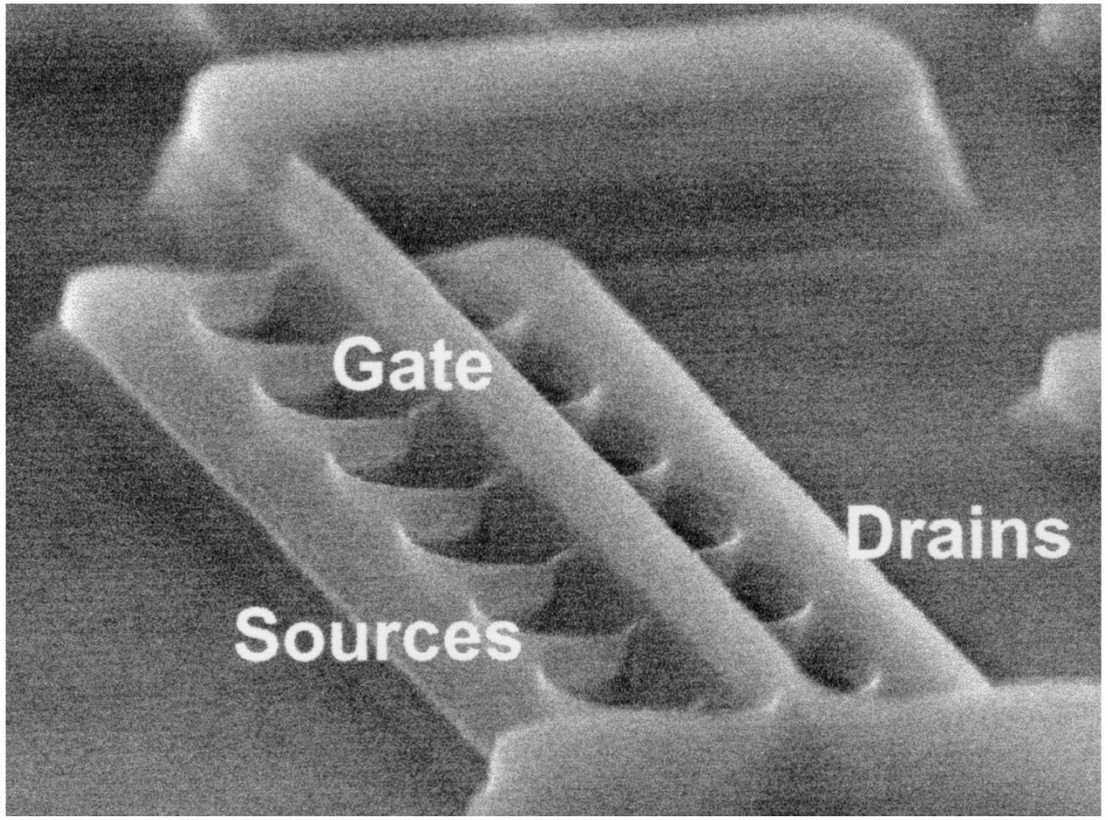


Intel 45 nm p-channel transistor (2007). Image: Intel; reproduced from W. A. Doolittle, Georgia Tech lecture slides (2011).

FinFET (22 nm, 2011): the gate wraps the channel



Planar (left) and 22 nm tri-gate (right). Source: Intel; reproduced from ETH Zürich lecture slides.



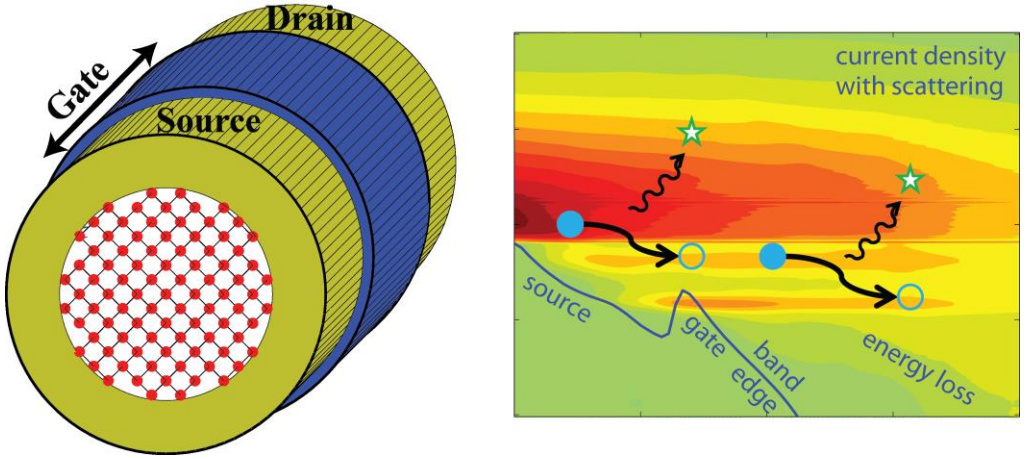
Early multi-fin FinFET: one gate across several fins. Source: Intel; reproduced from W. A. Doolittle, Georgia Tech lecture slides (2011).

The problem was now **electrostatic control**: the drain competes with the gate. Gating from three sides carried logic from 22 nm to 3 nm.

Gate-all-around nanosheets (2 nm class, 2025)

Foundry	Node	Architecture	Volume production
Samsung	3 nm	MBCFET (first GAA)	June 2022
TSMC	N2	GAA nanosheet	Q4 2025
Intel	18A	RibbonFET **+ PowerVia**	Announced Oct 2025
Samsung	SF2	GAA (MBCFET)	Q4 2025

Gate on **all four sides**; width set by the sheet, not in whole fins.
TSMC N2 vs N3E: +10–15% speed or −25–30% power.



Gate-all-around: the gate encloses the channel. reproduced from ETH Zürich lecture slides.

Next: backside power and CFET

Backside power delivery (2025 →)

About 15 metal layers carry both signals and power, and they compete for space. Move the **power network under the wafer**.

Intel
PowerVia on 18A (first, with GAA).
TSMC **Super Power Rail** on A16, volume H2 2026.

CFET (research; ~2032)

Stack the **nMOS on top of the pMOS** instead of side by side, removing the n-to-p spacing from the cell.

TSMC: working monolithic CFET inverter at 48 nm gate pitch. imec roadmap: production around
A5, ~2032. Treat as research.

6

Lithography: the enabling constraint

For twenty years, printing has gated everything else.

Printing below the wavelength

Architecture and lithography are coupled: when you cannot print smaller, **build in 3-D and let deposition set the critical dimension** (fin width, sheet thickness).

1

193 nm immersion (2004)

Water under the lens raises the numerical aperture: effective λ ≈ **134 nm**.

2

Multiple patterning (2010s)

Print, then print again on the offsets. Each pass costs a mask and overlay budget; cost per transistor stopped falling reliably.

3

EUV (2019)

13.5 nm light from laser-vaporised tin, all-mirror optics in vacuum, NA = 0.33.

4

High-NA EUV (now)

NA = **0.55**. Intel's first EXE:5200B accepted Dec 2025 for 14A; volume use 2027–28.

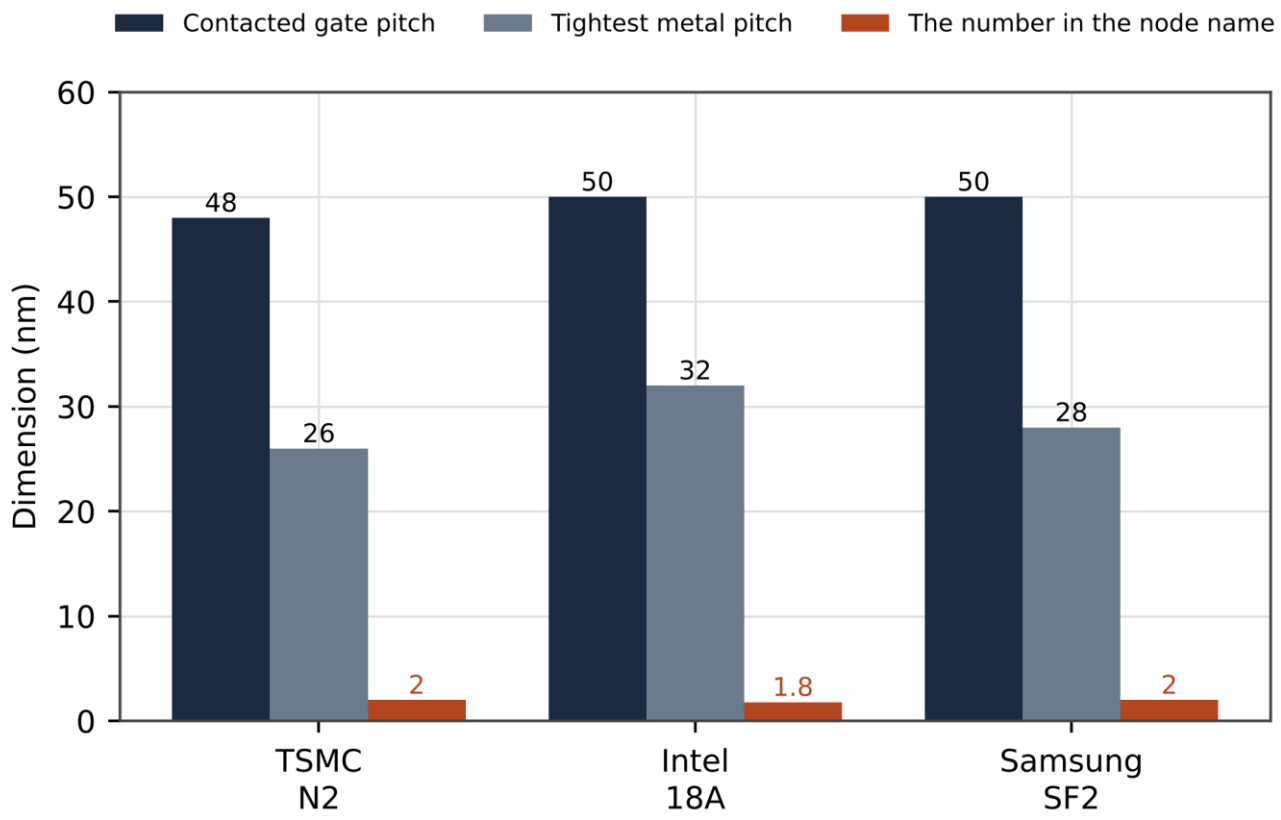
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The node-name problem

'2 nm' is not a length.

What the node name says, and what is on the wafer

- Node names were once the metal half-pitch; they are now **generation labels**
- The 2 nm processes in production:
  - TSMC N2: gate pitch **48 nm**, metal pitch 26 nm
  - Intel 18A: **50 nm**, 32 nm
  - Samsung SF2: **50 nm**, 28 nm
- Physical gate length: **~12 nm** (IBM 2 nm prototype, three stacked sheets)



The numbers to keep in your head

12–20 nm

Physical gate length. The number that makes the physics of Part 2 break.

45–50 nm

Contacted gate pitch of today's 2 nm-class processes.

“2 nm”

A label. Not a gate length, not a pitch.

8

The industry in 2026

Where you are likely to work.

Scale and structure

$792 bn

Global sales, **2025** (+25.6%, SIA)

~$1 tn

Projected for **2026** (WSTS)

~$300 bn

For comparison, **2010**

Foundries (TSMC) build others' designs · **Fabless** firms design only (Apple, NVIDIA, AMD, Qualcomm) · **IDMs** do both (Intel, Samsung) · **Equipment**: ASML's EUV monopoly is a strategic chokepoint. Growth is driven by **AI accelerators and high-bandwidth memory**.

Where this leaves us

Scaling continues, but nothing about it is free. The limits that bind are **device-physics limits**.

~10 nm

Mean free path of an electron in a Si inversion layer at 300 K

12–20 nm

Physical gate length in volume production today

~100 nm

Gate length around 2000: carriers scattered many times

A carrier crossing a modern channel may scatter **once, or not at all**. Drift-diffusion is at the edge of its validity: transport is **quasi-ballistic**. That is where Part 2 begins.

Figure credits

Reproduced figures

Photographs and micrographs: Bell Laboratories, Texas Instruments, Intel, IBM, Smithsonian Chip Collection, IEEE Spectrum (Oct 2002); G. E. Moore, Electronics (1965) and ISSCC (2003). Reproduced from W. A. Doolittle, Georgia Tech lecture slides (2011); S. Levkov, NJIT ECE 271 slides; ETH Zürich lecture slides. Copyright remains with the original owners.

Charts made for this course

Transistor count, clock frequency, technology timeline and node-name charts were generated for KKKC6353 from public data (Wikipedia; foundry announcements, accessed October 2026).